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- 1 The times taken by members of a large cycling club to complete a cross-country circuit have a normal distribution with mean μ minutes. The times taken, x minutes, are recorded for a random sample of 14 members of the club. The results are summarised as follows, where $\bar{x}$ is the sample mean.

$$\bar{x} = 42.8$$

$$\sum(x - \bar{x})^2 = 941.5$$

$$n \leq 15$$

Find a 95% confidence interval for μ .

[4]

$$C.I = \bar{x} \pm t_{\alpha/2, n-1} \sqrt{\frac{S_{xx}^2}{n}}$$

$$S_{xx}^2 = \frac{\sum(x - \bar{x})^2}{n-1} = \frac{941.5}{13} = 72.4$$

$$t_{\alpha/2, n-1} \rightarrow t_{0.025, 13}$$

$$\begin{aligned} & \downarrow \\ & 100(\alpha - 1) = 95 \\ & 2.16 \end{aligned}$$

$$42.8 \pm 2.16 \sqrt{\frac{72.4}{14}}$$

$$\rightarrow 42.8 \pm 2.16(2.27) = [37.9, 47.7]$$

- 2 A large number of students are taking a Physics course. They are assessed by a practical examination and a written examination. The marks out of 100 obtained by a random sample of 15 students in each of the examinations are as follows.

Student	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Practical examination	66	63	24	52	59	76	88	51	48	36	91	72	68	67	60
Written examination	63	57	39	50	47	71	87	65	56	39	78	70	61	62	70

Use a sign test, at the 10% significance level, to test whether, on average, the practical examination marks are higher than the written examination marks. [5]

use + / -

A → + B → + C → - D → + E → +
 F → + G → + H → - I → - J → -
 K → + L → + M → + N → + O → -
 10 + 5 -

$P(X \geq 10 | X \sim \text{Bin}(15, 0.5)) \rightarrow$ method for sign test

$P(X \geq 10 | X \sim \text{Bin}(15, 0.5))$

or $P(X \leq 5 | X \sim \text{Bin}(15, 0.5))$

$$P(X \geq 10) = \frac{15}{16} \times 0.5^{15} + \frac{15}{11} \times 0.5^{15} + \dots + \frac{15}{15} \times 0.5^{15}$$

$$= 0.151$$

H_0 : practical = written H_1 : practical > written

as $0.151 > 0.1 \rightarrow$ we reject $H_0 \rightarrow$ not in critical value critical region

- 3 A factory produces metal discs. The manager claims that the diameters of these discs have a median of 22.0 mm. The diameters, in mm, of a random sample of 12 discs produced by this factory are as follows.

22.4 20.9 22.8 21.5 23.2 22.9 23.9 21.7 19.8 23.6 22.6 23.0

- (a) Carry out a Wilcoxon signed-rank test, at the 10% significance level, to test whether there is any evidence against the manager's claim.

$$H_0: \text{med} = 22 \quad H_1: \text{med} \neq 22 \rightarrow \text{two tailed}^{[7]}$$

Value	Diff	Rank	P	N
22.4	0.4	2	2	
20.9	-1.1	8		8
22.8	0.8	5	5	
21.5	-0.5	3		3
23.2	1.2	9	9	
22.9	0.9	6	6	
23.9	1.9	11	11	
21.7	-0.3	1		1
19.8	-2.2	12		12
23.6	1.6	10	10	
22.6	0.6	4	4	
23.0	1	7	7	
			<u>54</u>	<u>24</u>

$$T = \min(P, N) = 24$$

critical value = 10%, two-tailed, $n = 12 \rightarrow 17$
 $24 > 17 \rightarrow \text{accept } H_0 \rightarrow \text{no skewness shown}$

insufficient evidence against manager's claim

- (b) State an assumption that is necessary for this test to be valid.

underlying distribution should be symmetric

- 4 The random variable Y is the sum of two independent observations of the random variable X . The probability generating function $G_Y(t)$ of Y is given by

$$G_Y(t) = \frac{t^2}{(4-3t)^4}.$$

- (a) Find $E(Y)$.

[3]

$$E(Y) = G'_Y(1)$$

$$G'_Y(t) = \frac{2t(4-3t)^{-4} + 12(4-3t)^{-5} \cdot t^2}{(4-3t)^8}$$

quotient rule

$$G'_Y(1) = 14$$

- (b) Write down an expression for the probability generating function of X .

[1]

$$G_X(t) = \frac{t}{(4-3t)^2}$$

(c) Find $P(X=4)$.

[3]

find t^4 term $\rightarrow t(4-3t)^{-2}$

$\hookrightarrow$ use binomial expansion

$$t(4)^{-2} \left(1 - \frac{3}{4}t\right)^{-2}$$

$$= \frac{t}{16} \left(1 - \frac{3}{4}t\right)^{-2}$$

$$= \frac{t}{16} \left(1 + \frac{3}{4}t + \frac{(-2)(-3)}{2!} \left(\frac{3}{4}t\right)^2\right.$$

$$\left. + \frac{(-2)(-3)(-4)}{3!} \left(\frac{3}{4}t\right)^3\right)$$

$$\frac{t}{16} \times -4 \times -\frac{27}{64} t^3 = \frac{27}{256} t^4$$

$$P(X=4) = \boxed{\frac{27}{256}}$$

- 5 Two companies, P and Q , produce a certain type of paint brush. An independent examiner rates the quality of the brushes produced as poor, satisfactory or good. He takes a random sample of brushes from each company. The examiner's ratings are summarised in the table.

Chi-square test

Company	Poor	Satisfactory	Good	
P	18	43	64	125
Q	22	22	31	75
	40	65	95	200

- (a) Test, at the 5% significance level, whether quality of brushes is independent of company. [7]

expected tabular values
 $= \text{Row total} \times \text{Column total}$

	Poor	Satisfactory	Good	
P	25	40.625	59.375	125
Q	15	24.375	35.625	75
	40	65	95	200

$$\text{test statistic} = \sum \frac{(\text{Obs} - \text{Exp})^2}{\text{Exp}}$$

$$= \frac{(25 - 18)^2}{25} + \frac{(40.625 - 43)^2}{40.625}$$

$$+ \frac{(59.375 - 64)^2}{59.375} + \frac{(15 - 22)^2}{15}$$

$$+ \frac{(24.375 - 22)^2}{24.375} + \frac{(35.625 - 31)^2}{35.625}$$

$$= 6.56$$

$$\begin{aligned}\text{degrees of freedom} &= (\text{row} - 1)(\text{col} - 1) \\ &= 1 \times 2 = 2\end{aligned}$$

$$\begin{aligned}\text{critical value} \\ &= \chi^2_{2}(0.95) = 5.991\end{aligned}$$

H_0 : Quality of brushes independent of company

H_1 : Quality not independent of company

as $6.56 > 5.991 \rightarrow \text{reject } H_0$

sufficient evidence to suggest that quality is not independent of company

(b) Compare the quality of the brushes produced by the two companies.

[1]

P produces better quality brushes than Q

- 6 Jade is a swimming instructor at a sports college. She claims that, as a result of an intensive training course, the mean time taken by students to swim 50 metres has reduced by more than 1 second. She chooses a random sample of 10 students. The times taken, in seconds, before and after the training course are recorded in the table.

Student	A	B	C	D	E	F	G	H	I	J
Time before course	54.2	47.4	52.1	59.0	55.3	51.0	48.9	52.2	58.4	51.4
Time after course	50.1	46.3	52.5	58.8	51.4	48.4	49.5	48.7	58.3	51.4

- (a) Test, at the 10% significance level, whether Jade's claim is justified.

[7]

t-test for paired differences

$$\sum d = 4.1 + 1.1 - 0.4 + 0.2 + 3.9 + 2.6 - 0.6 + 3.5 + 0.1 + 0 = 14.5$$

$$\sum d^2 = 4.1^2 + 1.1^2 + 0.4^2 + 0.2^2 + 3.9^2 + 2.6^2 + 3.5^2 + 0.1^2 = 52.81$$

$$s_d^2 = \frac{1}{n-1} \left(\sum d^2 - \frac{(\sum d)^2}{n} \right)$$

$$= \frac{1}{9} \left(52.81 - \frac{(14.5)^2}{10} \right) = \boxed{3.532}$$

Test stat = $\frac{\bar{d} - \mu_d}{\sqrt{\frac{s_d^2}{n}}}$

$\rightarrow \mu_b - \mu_a = 1 \rightarrow H_0$

$\mu_b - \mu_a > 1 \rightarrow H_1$

$$= \frac{1.45 - 1}{\sqrt{\frac{3.532}{10}}} = \frac{0.45}{\sqrt{3.532/10}} = 0.757$$

Critical value $\rightarrow$ one tailed, 10%, $n-1$
 $\rightarrow t_{0.9,9} = 1.383$

$0.757 < 1.383 \rightarrow$ accept H_0

$\rightarrow$ insufficient evidence to suggest that diff
in times is more than 1

(b) State an assumption that is necessary for this test to be valid.

[1]

underlying distribution of population
differences is normal

7 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{x}{4}(4-x^2) & 0 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find $\text{Var}(\sqrt{X})$.

$$\text{Var}(\sqrt{X}) = E((\sqrt{X})^2) - (E(\sqrt{X}))^2 \quad [4]$$

$$= E(X) - (E(\sqrt{X}))^2$$

$$\rightarrow \int_0^2 \frac{x^2}{4} (4-x^2) dx - \left(\int_0^2 \frac{x^{3/2}}{4} (4-x^2) dx \right)^2$$

$$\rightarrow \frac{1}{4} \left[\frac{x^3}{3} - \frac{1}{5} x^5 \right]_0^2 - \left(\frac{1}{4} \left[\frac{8}{5} x^{5/2} - \frac{2}{9} (x^2)^{3/2} \right]_0^2 \right)^2$$

$$\frac{16}{15} - \left(\frac{16}{9} \right)^2 = 0.0553$$

The continuous random variable Y is defined by $Y = X^2$.

- (b) Find the probability density function of Y .

[4]

$$F(x) = \int f(x) dx$$

$$= \frac{1}{9} \int_0^x 4t - t^3 dt = \frac{1}{4} \left[2t^2 - \frac{t^4}{4} \right]_0^x$$

$$= \frac{1}{16} (8x^2 - x^4)$$

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{16} (8x^2 - x^4) & 0 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$

$$y = x^2$$

$$F(y) = \begin{cases} 0 & y < 0 \\ \frac{1}{16} (8y - y^2) & 0 \leq y \leq 4 \\ 1 & y > 4 \end{cases}$$

$$\rightarrow 0 \leq y \leq 4$$

differentiate for $f(y) \rightarrow f(y) = \begin{cases} \frac{1}{16} (8 - 2y) \\ 0 \text{ otherwise} \end{cases}$

- (c) Find the exact value of the median of Y . $\rightarrow$ using C.D.F

[2]

$$\frac{1}{16} (8m - m^2) = 0.5$$

$$m^2 - 8m + 8 = 0$$

$$m = 4 - 2\sqrt{2} \rightarrow \text{as it is in the required range}$$

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